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A robust time-varying weight combined model for crude oil price forecasting

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ABSTRACT

Crude oil plays a vital role in industrial and social development and has become an integral part of the economic development. However, influenced by policies, wars, etc., it is hard to capture the trend of complex and volatile crude oil price if only one model is used, which often leads to poor forecasts. To enhance the prediction accuracy and robustness of forecasting models, a novel combined forecasting method with time-varying weights, i.e., Jaynes weight hybrid (**JWH**) model incorporating the Shannon information entropy and several forecasting methods is proposed in this paper. In the selection of the baseline models, the autoregressive integrated moving average in classical statistical forecasting strategies, back propagation neural network, extreme learning machine in neural network and long short-term memory neural network in deep learning models are chosen to fit the crude oil price. Four datasets and three experiments are constructed to verify the prediction ability of the novel combined forecasting model. Empirical results are calculated by five measurement criteria, suggesting that the prediction accuracy of the novel combined method is significantly higher than several comparison models and the mean absolute percentage error of the model has arrived 2.81 % in detail. Particularly, the proposed model has achieved satisfactory performances in Covid-19 and the War in Ukraine, further verifying the robustness of the combined methodology.

List of abbreviations.

AR	autoregressive
MA	moving average
ARIMA	autoregressive integrated moving average
ANN	artificial neural network
FNN	feedforward neural network
ELM	extreme learning machine
BPNN	back propagation neural network
EWH	equal weight hybrid
JWH	Jaynes weight hybrid
MAE	mean absolute error
RNN	recurrent neural network
GRU	gate recurrent unit
LSTM	long short-term memory
SOA	seasons optimization algorithm
MLR	multiple linear regression
EMD	empirical mode decomposition
EEMD	ensemble empirical mode decomposition
MAPE	mean absolute percentage error
sMAPE	symmetric mean absolute percentage error
MSE	mean squared error
RMSE	root mean squared error
List of symbols.	
y_t	observed value
$\hat{y}_t$	predicted value
z_t	output of BPNN
w_i	weight of i-th model
w_{it}	weight of model i-th at t
X	input matrix of network
Y	output matrix of network
σ	activation function
T	target matrix of ELM

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(*continued*)

C	regularization parameter of ELM
β^*	best solution of ELM
f_t	forgetting gate of LSTM
i_t	input gate of LSTM
o_t	output gate of LSTM
c_t	internal state of LSTM
$\widetilde{c}_t$	candidate state of LSTM
h_t	external state of LSTM
$\theta_1,\theta_2,\ldots,\theta_p$	autocorrelation coefficients
$\beta_1,\beta_2,\ldots,\beta_q$	moving average coefficients
ε_t	random white-noise
p	order of AR model
q	order of moving average model
d	order of the difference
x	input vector of network
f	output vector of network
W,U,V,w_{ij},β	weight of network
B,b	bias of network
L	number of nodes on hidden layer
M	number of nodes on output layer
m	number of single models
n	number of samples
$H(t)$	Shannon information entropy
$\odot$	operation of multiplying the vector elements

1. Background

Known as the blood of industry, crude oil is an indispensable role in modern industry. For example, gasoline, kerosene, diesel oil and natural gas refined from crude oil have become an indispensable necessity in civil life [1]. The production process of materials such as synthetic and organic materials is inseparable from crude oil, so it can be said to be a blood-like presence in the chemical industry, which directly affects the overall economic [2].

The cost of energy can be directly affected by crude oil price, then consumption and investment decisions are influenced in turn. It is also critical to decision-making such as oil production, refining and energy investment, benefiting for energy companies to develop strategies and plan investments to ensure the sustainability and profitability of their businesses [3]. In addition, the trend of crude oil price is influenced by international political and geopolitical factors, and forecasting crude oil price is beneficial for national and international organizations to understand the international situation and make reasonable plans before some emergencies, so as to alleviate the national energy security problems [4]. Therefore, crude oil price forecasting have become the focus of central banks, governments and energy companies [2,5].

Due to its large demand and supply and the characteristic that it always be heavily influenced by the global economy and policy, crude oil is one of the most widely known commodities and its original price often deviates from the normal market price range, showing obvious uncertainty, nonlinearity, mutation and instability, which brings a great difficulty to the crude oil price forecasting. Especially, in the face of emergencies, such as Covid-19, the trend of crude oil price is much more volatile [6].

The context of this research is given before the start of the study. **Section 2** provides the literature review of the crude oil price forecasting, introduces the motivation and contributions of this paper based on the existing studies. **Section 3** describes the detailed methods adopted in this research. **Section 4** introduces the cases, evaluation metrics and empirical results. The discussion and conclusion are summarized in Sections 5 and 6, respectively.

2. Literature review

Thanks for the significance of crude oil price forecasting, domestic and foreign research scholars and experts have conducted a large number of works on crude oil price forecasting. There are many relatively mature forecasting models, especially in Europe and the United States, for example, Drachal [7] proposed an averaging time-varying VAR models, Abdollahi [8] constituted a hybrid model by combining ensemble decomposition methods, optimization algorithms and machine learning models, and Rubaszek [9] used DSGE model to forecast crude oil price.

Silva et al. [10] indicated that compared with the fundamentalist approach, time series models are conveniently for capture and analyze the price's past behavior and predict future movements, therefore, time series forecasting methodologies are used to predict crude oil price in this paper. Analyzing the articles published in recent years, the time series models used for crude oil price forecasting mainly involved traditional statistical and econometric models, machine learning models and combined models [11].

2.1. Traditional statistical and econometric models

Classical statistical and econometric forecasting models include the autoregressive integrated moving average (ARIMA) [12–16], generalized autoregressive conditional heteroscedasticity type models (GARCH-type) [1,17–20], grey forecasting models [21–23], and so on.

2.2. Machine learning models

There are various factors such as international environment and emergency cause the fluctuation of crude oil price, which showing obvious uncertainty, nonlinearity, mutation and instability. Therefore, traditional methods with strong linear fitting ability are difficult to obtain satisfactory results. In contrast, machine learning methods including neural network methods and deep learning methods, have great advantages to extract nonlinear characteristics of time sequences. Therefore, many scholars have introduced machine learning methods for crude oil price prediction.

Back propagation neural network (BPNN) is the simplest neural network providing general computational power to process numerical data and learn a nonlinear mapping between inputs and outputs. Zhang et al. [24] developed a sales prediction model based on BPNN, principal component analysis and optimization algorithm with historical sales data, sentiment index and other variables, which effectively improved the prediction performance. However, a fatal drawback of BPNN is its slow convergence and gradient vanishing, mainly because its gradient descent dynamics [25].

Extreme learning machine (ELM) can be considered as the improved version of feedforward neural networks and back propagation algorithms with advantages in learning rates and generalization capabilities, thus it is widely applied to time series forecasting in many areas, such as energy, environment and economy [26]. For example, Weng et al. [27] developed a genetic algorithm regularization online extreme learning machine with a forgetting factor to forecast the volatility of crude oil futures, Zhang et al. [28] applied ELM and meta-heuristic algorithms to predict copper price, Wang et al. [29] combined ELM and empirical mode decomposition (EMD) to construct a streamflow forecasting model. However, the best solution can be affected by the initial input weights and bias, leading a risk of falling into local minima.

Long short-term memory neural network (LSTM) has also been introduced to time series forecasting over the past few years. Recurrent neural network (RNN) is an effective method for time series prediction because it can mine the temporal and semantic information in the data [30]. LSTM as a special type of RNN is also frequently applied in many studies of time series prediction. For instance, the footprints of RNN and its variants have already appeared in energy and financial forecasting, including solar irradiance prediction [31], and cryptocurrency prices prediction [32], etc. LSTM has been used to stock prices forecasting [33], metal price forecasting [34], photovoltaic power prediction [35] and energy demand forecasting [36]. Unfortunately, although gates benefit to overcome the gradient disappearance, important data still be lost in LSTM when the input sequence is too long [37].

2.3. Combined models

Among different prediction problems, scholars usually choose appropriate methods in accordance with the structural features of given data, such as linearity or nonlinearity. However, in a complex real world, the prediction problems influenced by various factors are often complex, and single model is hard to explain or predict the complex situations well. For the sake of enhancing the performance of forecasting models, the combined model method was suggested [38]. Multiple models with the same or different structures are selected for predicting the same time series. During the prediction process, the corresponding weights are set according to the calculated prediction errors, and the ultimate forecasting sequence can be obtained through weighting the forecasting sequence of these models. The first combination of different methods was proposed in 1969 [39]. To date, in addition to the classical economic prediction methods, more and more machine learning methods have been proposed to provide additional elements for hybrid models with good performance. For example, Safari and Davallou [40] proposed a Kalman filter-based method for determining weights that can perform better than the combined method with equal weights and the combined model based on genetic algorithm [41].

In addition to using different optimization algorithms for weights calculation to build combined models, model stacking or integration are used to design forecasting models in many studies, combining the advantages of each single model. For example, Jiang et al. [42] created a decomposition ensemble crude oil price prediction model combining machine learning approaches, including ensemble EMD (EEMD), seasons optimization algorithm (SOA), gate recurrent unit (GRU) and multiple linear regression (MLR). Wang et al. [43] developed an enhanced hybrid model based on multiple influencing factors and divide-conquer strategy for carbon price prediction. Che et al. [44] proposed a novel ultra-short-term probabilistic wind power forecasting with spatial–temporal multi-scale features and K-FSDW based weight. Hao et al. [45] designed a bi—level ensemble learning approach by combining decomposition—ensemble forecasting and resample strategies.

Several researches on crude oil price prediction in recent years are summarized in Table 1 and we can obtain two comments through the study of historical literature.

Comment 1: In fact, whether traditional modes, deep learning methods, none is universal and always achieves the best prediction performance in all cases. In particular, traditional statistical methods are more suitable for linear time series prediction while perform poorly on nonlinear datasets, and although machine learning methods can fit nonlinear characteristics of time series while their prediction results often depend on the selection of hyperparameters.

Comment 2: Combined forecasting model is a strategy that can significantly improve the prediction performance, constant weighted model is one of the mature manifestations of it, which ignores the variability of model across the different data points. Giving each model a variable weight is more beneficial for achieving higher accuracy than giving each model a constant weight.

2.4. Motivation of this study

As introduced in the literature review, there has been many studies in the field of crude oil price forecasting. The combined method has more advantages compared with the single model prediction, which is manifested in the improvement of the prediction accuracy and applicability. Therefore, the combined model has received more and more attention. However, the relevant studies of the combined method focus on the constant weighted models, the superiority in combining the advantages of each model is very limited. Besides, crude oil price is more susceptible to various complex factors, showing strong nonlinearity, volatility and instability, leading the necessary to design a prediction model that is good at comprehensive analysis of nonlinear data and linear data. Therefore, a time-varying weight combined model is proposed for crude oil price sequence to solve its forecasting problem, optimizing from the following two aspects.

1) Combined methods are already common in crude oil price forecasting, however, in most studies, constant weights have be given to individual models, ignoring the uncertainty of weights. Therefore, both the forecasting error and the uncertainty of the weights should be taken into consideration simultaneously during the establishment of the combined model. Theoretically, taking prediction error and the uncertainty of weights to model by time-varying weights can effectively improve the prediction accuracy.
2) Establishing multiple weight combined strategies and analyzing the stability of the combined method are attempted to existing studies rarely. In fact, a model with poor prediction effect can easily cause perturbations in the performance of the combined method, causing the inadequate robustness of the combined method. Therefore, it is necessary to establish a combined strategy who is not susceptible to poor performance models with good prediction accuracy, high robustness.

Table 1
Recent studies for crude oil price forecasting.

Category	Authors	Models	Case	Horizon	Performance	Feature/contribution
Traditional models	Naderi et al. [14]	ARIMA	The Central Bank of the Islamic Republic of Iran	Monthly; daily; annual	The RMSE values of ARIMA based monthly, daily, annual oil price are 3.440, 0.071, 4.261, respectively.	In this research, a univariate ARIMA is used to predict oil price.
	Lin et al. [17]	GARCH; HM-GARCH	WTI crude oil price; Chinese Daqing crude oil price	Daily	The performance of the proposed HM-GARCH model is better than other regular GARCH-type models.	In this paper, several GARCH-type models are compared in oil price forecasting.
Machine learning models	Kristjanpoller and Minutolo [19]	ANN - GARCH	Crude oil spot price; oil futures price from Bloomberg	Daily	The MAPE of ANN-GARCH is 0.4021 % with a 29.79 % reduction.	By combining ANN and GARCH, this study achieves a significant improvement in forecasting accuracy.
	Wang et al. [29]	ELM	Crude oil futures prices from the US EIA.	Monthly	The MAPE with three steps is 1.7869 %.	The impact of internet attention on crude oil markets is added to establish effective model.
Combined models	Safari and Davallou [40]	EWH, GWH, PHM and ZHM	OPEC crude oil price; WTI crude oil price	Monthly	The MAPE of PHM are 9.54 % and 8.31 % using a relative ratio of 90 % and 80 % on the testing data of OPEC, and 2.44 % on the testing data of WTI.	In this article, time-varying weights using the Kalman-filter is employed.
	Jiang et al. [42]	EEMD-SOA-GRU-MLR	WTI crude oil price	Daily	The proposed model has the best forecasting performance in all cases.	A decomposition-ensemble model based on SOA, EEMD, GRU and MLR is proposed.

2.5. Primary works and contributions

Since time series data usually has both linear and nonlinear structures, traditional statistical models that are better at capturing linear features and machine learning models that can handle nonlinear dependence are used to predict and some weights are given to them, which is more conducive to obtain higher prediction accuracy. In this study, both the forecasting error and the uncertainty of the weights are taken into consideration simultaneously during the establishment of the combined model. Particularly, Shannon information entropy proposed by Shannon in 1949 is introduced to measure the uncertainty of the weights, and Jaynes maximum entropy [46] is chosen to achieve the maximum of Shannon information entropy. The contribution of different models to the combined model at different time slices is represented by a set of time-varying weights, equivalent to a random variable with uncertainty, which can be well described by the Shannon information entropy. Thus, the information entropy is used to calculate the variable weights of different models in this paper. As a result, a combined forecasting model integrating Jaynes maximum entropy and minimum quadratic loss is proposed, which is established by ARIMA, BPNN, ELM and LSTM.

Two international crude oil price benchmarks, WTI and Brent datasets are selected to design experiments to compare the forecasting ability of this novel combined model and several comparison models. Energy systems worldwide are significantly impacted by post Covid-19 pandemic and the War in Ukraine [47,48], which also causes the huge fluctuations of crude oil price, increasing the difficulty for its prediction. Empirical results provide evidence for the better effect of the proposed model, especially during Covid-19 and the War in Ukraine, which further validates the robustness and stability of the proposed method.

Overall, the contributions of proposed model can be summarized as follows.

1) A novel combined strategy is proposed and improve the prediction accuracy significantly. A time-varying weight combined strategy is proposed for crude oil price forecasting, which is based on mathematical theory, improving the performance and providing a reference for the actual crude oil price forecasting.
2) Enhance the stability of the model and make it less affected by the models with poor performance. To combine the advantages of these single models, we selected four representative models and tried two strategies with three model combinations and four models combinations, showing that the performance of the latter is slightly better, indicating that the new combined method is less affected by poorly performing models.
3) Improve the robustness of the model to enhance the performance of the model during special periods. The proposed combined model has good robustness. Specifically, for the highly volatile data during post Covid-19 pandemic and the War in Ukraine, the performance of the proposed model still relatively better, reflecting its universality.

3. Methodology

3.1. Elementary forecasting models

Three types of classical models: ARIMA in classical statistical method, BPNN and ELM in classical neural network method, and LSTM in deep learning method used in this research are introduced in detail.

3.1.1. Autoregressive integrated moving average model (ARIMA)

As one of the traditional prediction and analysis methods in statistics, the core thought behind ARIMA is an assumption introducing a random error, i.e., there is a linear dependence between future values and past observations with a random error [40].

A complete ARIMA can be split into three parts, including AR (p), I (d), and MA (q), whose meaning and mathematical expression are declared as follows.

AR (p) represents the autoregressive model that indicates the lags in a stationary process as the following equation shows:

$$\widehat{y}_t = \mu + \theta_1 y_{t-1} + \theta_2 y_{t-2} + \ldots + \theta_p y_{t-p} + \varepsilon_t \tag{1}$$

where p is the order of AR model, $y_{t-1}, y_{t-2}, \ldots, y_{t-p}$ are historical observed sequence, $\theta_1, \theta_2, \ldots, \theta_p$ are autocorrelation coefficients, ε_t is the random white-noise.

I (d) denotes the difference process, where d is the order of the difference. To establish a measurement model of a non-stationary time series, the first step is to convert it into a stable sequence through difference, d represents the order of this difference.

MA (q) is moving average model that shows lags in random error. When order is equivalent to q, the moving average can be formed as follows:

$$\widehat{y}_t = \mu + \beta_1 \varepsilon_{t-1} + \beta_2 \varepsilon_{t-2} + \ldots + \beta_q \varepsilon_{t-q} + \varepsilon_t \tag{2}$$

where q is the order of moving average model, $\beta_1, \beta_2, \ldots, \beta_q$ are moving average coefficients, ε_t is random white-noise.

ARIMA is one of the simplest models of time series prediction, which only asks for endogenous variables without needs for any other exogenous variables. However, for the reason that ARIMA essentially can only captures linear correlations, the nonlinear dependence and structures are not used to modeling.

3.1.2. Back propagation neural network (BPNN)

As the most basic error propagation algorithm, BP algorithm is the most core and essential part of the feedforward neural network, and BPNN constructed from it is frequently applied to regression and classification cases.

The training procedure of BPNN can be considered as a procedure to solve a nonlinear unconstrained extreme problem with given learning samples. The weights and bias can be adjusted through backward propagation until achieve the minimum error square, namely:

$$\min E = \frac{1}{2}\sum\nolimits_{t=1}^{n}(y_t - z_t)^2 \tag{3}$$

where y_t is the observed value and z_t is the output of BPNN, which could be calculated through the following formulas:

$$s_j = \sigma\left(\sum\nolimits_{i=1}^{N} w_{ij} x_i + b_j\right) \tag{4}$$

$$z_k = \sigma\left(\sum\nolimits_{j=1}^{J} w_{jk} s_j + b_k\right) \tag{5}$$

where σ is an activation function defaulting to sigmoid, x is input vector, s_j and z_k denote the outputs of hidden layers and the output layer, σ and b_j represent weight and bias of hidden layer while w_{jk} and b_k represent weight and bias of output layer.

Several gradient-based algorithms such as Stochastic Gradient Descent and Momentum algorithm can be used in the update process of weights, which makes it possible to sink into the local minimum points. Besides, the training speed is very slow, and the algorithm does not necessarily converge.

3.1.3. Extreme learning machine (ELM)

ELM is an effective feedforward neural network with one hidden layer which is developed by Huang et al. [49]. Different from classical single-hidden layer feedforward neural network, ELM starts with random selection of weights and bias of its hidden layer, and the output layer weights are accessed directly by optimizing a minimization problem formed by learning error section and regular of the output layer weight norm according to the Moore-Penrose generalized inverse method [50]. The output of ELM and the optimization objective of ELM can be described as given below:

$$\min_{\beta \in \mathbb{R}^{L \times M}} \frac{\beta^2}{2} + \frac{C}{2} H\beta - T^2 \tag{6}$$

$$Y(x) = H\beta = \sum\nolimits_{i=1}^{L} h_i(x)\beta_i \tag{7}$$

$$H = \sigma(W \cdot X + B) \tag{8}$$

$$\sigma(x) = \frac{1}{1 + e^{-x}} \tag{9}$$

where X and Y denote the input and output, respectively; β is the weight matrix of output layer, W and B are input weight and bias, respectively. L and M denote the size of hidden layer and output layer, respectively. $\sigma(x)$ represents the sigmoid function. T represents the objective matrix of training samples, C denotes the regularization coefficient. Based on MoorePenrose generalized inverse method, the best solution could be computed as:

$$\beta^* = H^T \left(\frac{1}{C} + HH^T \right) T \tag{10}$$

In contrast to the gradient-based iterative adjustment algorithm, ELM retains the universal optimization power of SLFN, and it also has advantages in learning parameters, training speed, and generalization ability [29], which enable it to be used more frequently in neural networks. However, as shown in formula (10), the best solution can be affected by the initial input weights and bias, leading to the acquisition of a locally optimal solution.

3.1.4. *Long short-term memory (LSTM)*

LSTM, developed by Hochreiter and Schmidhuber [51], is a widely used variant of RNN that is one of the most frequently used models developed to dig for dependencies between sequences [52]. The reason why the performance of RNN is excellent on time series forecasting is that the input of the current time slice includes the output of the middle layer at the former time slice, which provides an approach to take the previous information to calculate the content of the current time slice, shown in Fig. 1. The calculation process of RNN could be given as Eqs. (11) and (12):

$$h_t = \sigma(Vh_{t-1} + Ux_t + b) \tag{11}$$

$$m_t = W \cdot h_t \tag{12}$$

where x_t and m_t represent the input and output of the model respectively and U denotes the weight matrix of the input; h_{t-1} is model internal short term memory and V_h denotes the weight matrix of it; σ denotes the sigmoid activation, W is the weight of the output layer.

As we can see from Fig. 1 and formula (11), in RNN, the information of the hidden layer at this time slice only be calculated by the current input and the information of the hidden layer at previous time, leading that the stored information is usually short-term information. When a long-term dependence can be found from the data, the problem of gradient disappearance of RNN may appear, which is the reason why

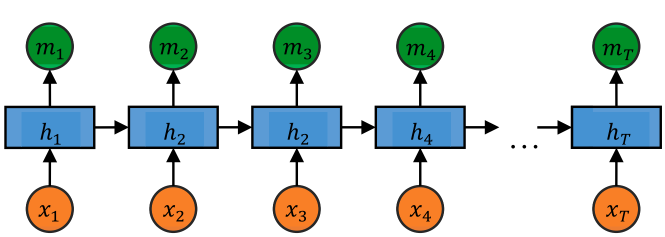

Fig. 1. The structure of a general RNN.

LSTM is invented. Compared to the general RNN, gate mechanism and cell state are important components of neurons, involving in information transmission.

The schematic diagram of LSTM is depicted in Fig. 2, and the mathematical dynamics can be described as Eqs. (13)–(18):

$$f_t = \sigma\left(W_f x_t + U_f h_{t-1} + b_f\right) \tag{13}$$

$$i_t = \sigma(W_i x_t + U_i h_{t-1} + b_i) \tag{14}$$

$$o_t = \sigma(W_o x_t + U_o h_{t-1} + b_o) \tag{15}$$

$$c_t = f_t \odot c_{t-1} + i_t \odot \widetilde{c}_t \tag{16}$$

$$h_t = o_t \odot \tanh(c_t) \tag{17}$$

$$\widetilde{c}_t = \tanh(W_c x_t + U_c h_{t-1} + b_c) \tag{18}$$

where f_t, i_t and o_t represent three gates, i.e., forgetting, input and output, used to control information forgetting, preservation and output respectively; c_t is internal state, a new variable compared to the general RNN; h_t is external state of the hidden layer; $\widetilde{c}_t$ is candidate state, an intermediate variable used to calculate the internal state c_t; W and U are weight matrix; b is the bias of the corresponding calculation steps; σ is sigmoid activation function; $\odot$ denotes the operation of multiplying the vector elements.

3.2. *Combined model*

3.2.1. *Theory of combined forecasting model*

Combined prediction method first proposed by Bates and Granger [39] is designed to answer the question that a single forecasting model is susceptible to environmental factors. At each step of prediction, we use single models to forecast and weight these results to obtain the final result. For a certain prediction problem, a total of m single models are selected. We assume that in the period t, $\widehat{y}_t$ is the results of model, $\widehat{y}_{it}$ is the forecasting value of model i, $w_{it} > 0$ is the weight of model i, then the mathematical expression of the combined model is [41]:

$$\widehat{y}_t = \sum\nolimits_{i=1}^{m} w_{it} \widehat{y}_{it}, \sum\nolimits_{i=1}^{m} w_{it} = 1 \tag{19}$$

The principle of reasonably determining the weights is to minimize the sum of errors. Here minimizing the sum of error squares is chosen as the optimization objective, namely:

$$\begin{cases} \min \sum\nolimits_{t=1}^{n} \left(y_t - \sum\nolimits_{i=1}^{m} w_{it} \widehat{y}_{it} \right)^2 \\ \quad s.t. \quad \sum\nolimits_{i=1}^{m} w_{it} = 1 \end{cases} \tag{20}$$

In this study, several methods to determinate the weights of single models used would be introduced respectively in subsection 3.2.2.

3.2.2. *Calculating the weights in the combined model*

Combining different prediction models with appropriate weights to get stable and effective prediction results is the principle of combined models. One of the most important steps is to determine the weights of selected methods. Two approaches are designed in this paper to calculate these.

3.2.2.1. Constant (equal) weights. One of the most common methods to establish combined model is equal weight hybrid model (**EWH**), which determinate the weights through simple averaging. Besides, the weights are constants, namely:

$$w_{it} = \frac{1}{m}, i = 1, 2, \ldots, m, t = 1, 2, \ldots, n \tag{21}$$

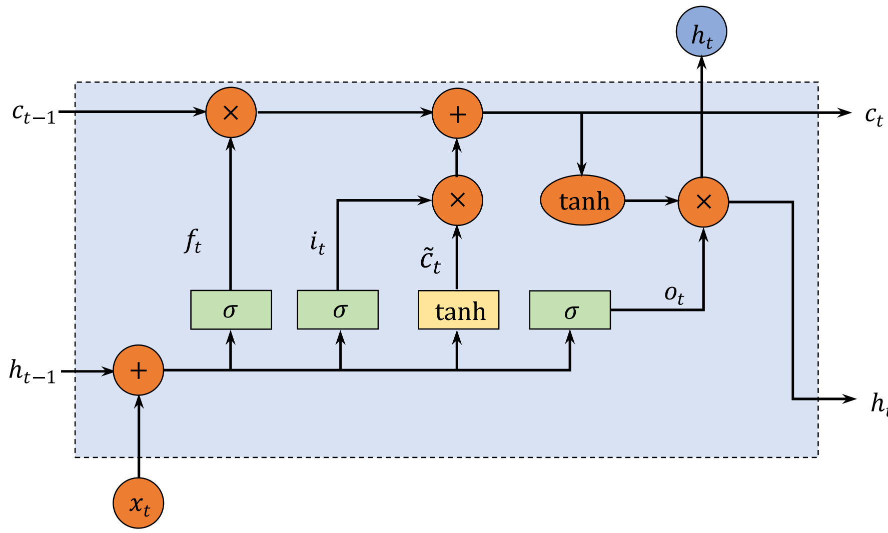

Fig. 2. The structure of LSTM [35].

3.2.2.2. *Time-varying (Jaynes) weights.* Since the weights of single models at different moments is a random variable, i.e., with uncertainty. Through taking the uncertainty of the weight into consideration, the Shannon information entropy is used to establish the Jaynes hybrid model (**JWH**). The Shannon information entropy can be expressed:

$$H(t) = -\sum_{i=1}^{m} w_{it} \ln(w_{it}) \tag{22}$$

According to the principle of Jaynes maximum entropy, a reasonable weight should make Shannon information entropy maximum, that is, the following formula:

$$\begin{cases} \max H(t) = -\sum_{i=1}^{m} w_{it} \ln(w_{it}) \\ \quad s.t. \sum_{i=1}^{m} w_{it} = 1 \end{cases} \tag{23}$$

Combining the above two objectives of Eqs. (20) and (23), the problem of solving the optimal weight is equivalent to solving a combined optimization problem:

$$\begin{cases} \min \beta \sum_{t=1}^{n} \left(y_t - \sum_{i=1}^{m} w_{it} \widehat{y}_{it} \right)^2 + (1-\beta) \sum_{i=1}^{m} w_{it} \ln(w_{it}) \\ \qquad s.t. \sum_{i=1}^{m} w_{it} = 1 \end{cases} \tag{24}$$

Specifically, the parameter β $(0 \le \beta \le 1)$ is the balance coefficient between the two target representations which can be given in advance based on the actual problem.

To calculate the weights for each models, by the Lagrange multiplier method, the problem (24) can be convert to the following unconstrained question:

$$L(w_{it}, \beta, \lambda) = \beta \sum_{t=1}^{n} \left(y_t - \sum_{i=1}^{m} w_{it} \widehat{y}_{it} \right)^2 + (1-\beta) \sum_{i=1}^{m} w_{it} \ln(w_{it}) + \lambda \left(1 - \sum_{i=1}^{m} w_{it} \right) \tag{25}$$

Then, the gradient of w_{it} is calculated, namely:

$$\frac{\partial L}{\partial w_{it}} = \beta \frac{\partial \sum_{t=1}^{n} \left(y_t - \sum_{i=1}^{m} w_{it} \widehat{y}_{it} \right)^2}{\partial w_{it}} + (1-\beta)(1 + \ln(w_{it})) - \lambda \tag{26}$$

In this case, the error can be changed from formula (3) to:

$$E = \frac{1}{2} \sum_{t=1}^{n} \left(y_t - \sum_{i=1}^{m} w_{it} \widehat{y}_{it} \right)^2 \tag{27}$$

The derivative of E with respect to w_{it} is:

$$\frac{\partial E}{\partial w_{it}} = \frac{\partial E}{\partial \widehat{y}_t} \frac{\partial \widehat{y}_t}{\partial w_{it}} = (\widehat{y}_t - y_t) \widehat{y}_{it} \tag{28}$$

Therefore,

$$\frac{\partial L}{\partial w_{it}} = 2\beta (\widehat{y}_t - y_t) \widehat{y}_{it} + (1-\beta)(1 + \ln(w_{it})) - \lambda \tag{29}$$

In the same way, the gradient of β, λ is calculated separately, namely:

$$\frac{\partial L}{\partial \beta} = 2E - \sum_{i=1}^{m} w_{it} \ln(w_{it}) \tag{30}$$

$$\frac{\partial L}{\partial \lambda} = 1 - \sum_{i=1}^{m} w_{it} \tag{31}$$

In the initialization phase of the algorithm, we set $w_{it} = \frac{1}{m}$, $\beta = \frac{1}{2}$, and $\lambda = \frac{1}{2}$. During training or forecasting, these three parameters can be optimized synchronously based on the results of the function $L(w_{it}, \beta, \lambda)$.

3.3. *The proposed combined model*

The crude oil price forecasting framework in this research is illustrated in Fig. 3, consisting of five parts, namely, data collection, datasets division, benchmark models, combined models, evaluation criteria and performance analysis.

1) Data collection (as described in Fig. 3 (a)): There are four datasets constructed in this study, using two international benchmarks, WTI and Brent, which are collected with the frequency of weekly or monthly. The detailed introduction and their statistical features will be confirmed in subsection 4.2.
2) Datasets division (as described in Fig. 3 (b)): Regardless of the number of samples, the same percentage partition is used to divide the datasets. The training samples is defined as the first 80 % of the dataset, while the remaining 20 % is deemed to the testing samples.
3) Benchmark models (as described in Fig. 3 (c)): In order to establish the combined model, integrate the characteristics of classical models and deep learning models, four single models are applied to predict crude oil price, including LSTM, ELM, ARIMA, BPNN.

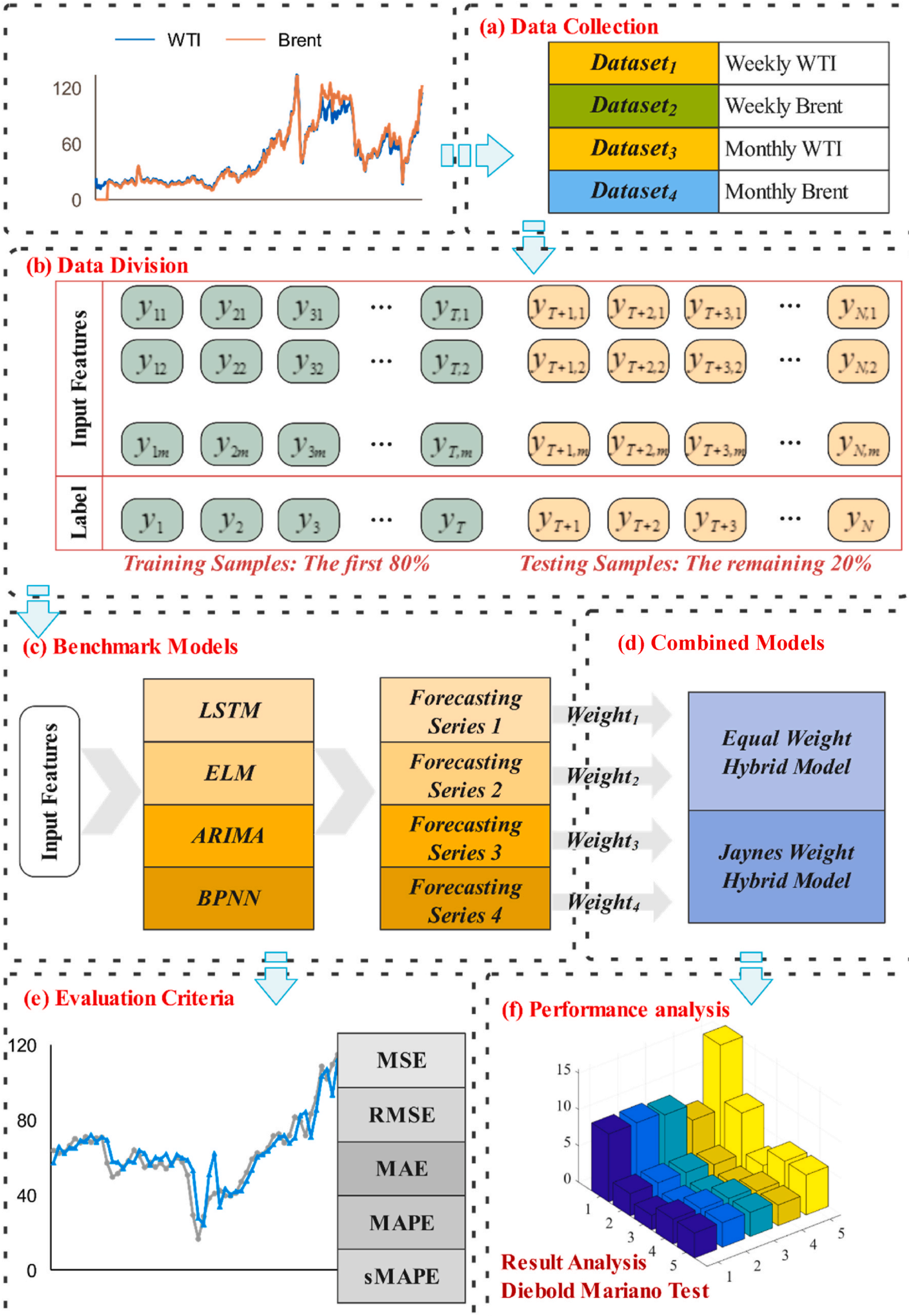

Fig. 3. The overall framework of crude oil price forecasting.

4) Combined models (as described in Fig. 3 (d)): In this paper, equal weights and weights calculated by Jaynes maximum entropy are designed to establish combined models.
5) Evaluation criteria (as described in Fig. 3 (e)): To compare the elementary models from a digital perspective, five error metrics are employed to measure the performance of several methods.
6) Performance analysis (as described in Fig. 3 (f)): For the reason that the overall fitting performance of each model to the testing samples is also one of the indispensable measurement indicators, the contrast curves of the observed value and the predicted value and the error curves would be plotted in this part.

4. Experiments and results analysis

The accurate forecasts of crude oil price is significantly important for the stability of national economic and the prosperity of oil enterprises. Therefore, a time-varying combined forecasting model has been established to get prediction results with high accuracy. To verify the performance of the proposed model, three experiments are designed in this section and the description about the design of experiments, the introduction of datasets, the selection of measurement criteria, and the design details of three experiments are shown in the following subsections.

4.1. Experiments design

For the purpose to validate the effectiveness and applicability of the proposed model, three practical cases are analyzed in depth. In the following experiments, weekly and monthly sequences of two international crude oil price benchmarks, i.e., West Texas Intermediate (WTI) benchmark and Brent benchmark are selected to build the datasets. As for the forecasting models, four benchmark models including LSTM, ELM, BPNN and ARIMA, Jaynes weight combined method based on four models (**JWH**), equal weight combined model based on four models (**EWH**), Jaynes weight combined method based on three models (**JWH$_3$**), equal weight combined model based on three models (**EWH$_3$**) are selected to make comparisons. Note that the combined models based on these three models are named as **JWH$_3$** and **EWH$_3$** in the remaining description of the paper to distinguish from the combined models based on the four benchmark models.

In ***Experiment I,*** the weekly WTI and Brent crude oil price datasets and five evaluation indicators are used to compare the prediction accuracy of the proposed **JWH** with **EWH** and four benchmark models, i.e., ARIMA, ELM, BPNN, and LSTM. In order to further reveal the validity and applicability of **JWH**, ***Experiment II*** is designed to validate the effectiveness of it in the monthly crude oil price forecasting. Furthermore, two new datasets obtained by splitting WTI weekly price dataset are added to ***Experiment III*** as the final explanation of the applicability of this combined method.

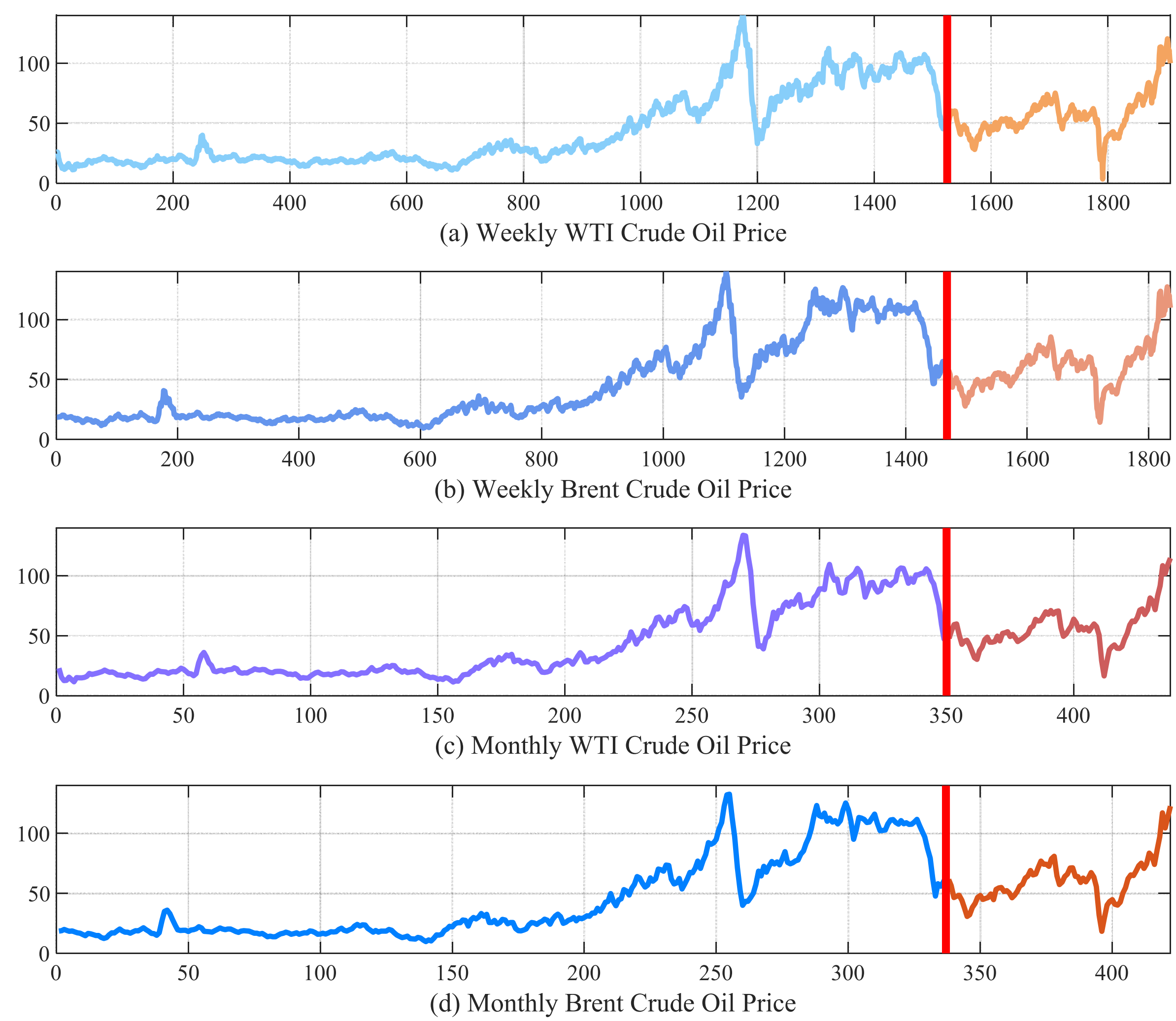

Fig. 4. Overall trends of crude oil price series.

4.2. Datasets

In this work, two international crude oil price benchmarks, including West Texas Intermediate (WTI) benchmark and Brent benchmark are selected to design experiments. Weekly data of the two benchmarks are named **Dataset$_1$** and **Dataset$_2$**, similarly, monthly data of the two benchmarks are named **Dataset$_3$** and **Dataset$_4$**, separately. Moreover, these datasets can be downloaded from U.S. Energy Information Administration (https://www.eia.gov/).

Datasets used in this study are first collected at or after 1986, and the fluctuation of crude oil price up to now can be depicted in Fig. 4. Oil prices from 1986 to 2022 can be roughly divided into six periods as shown in Table 2.

Influenced by market supply and demand, geopolitics, inflation, economic situation and other factors, crude oil price are not stable, but overall upward trends can be portrayed. In 2020, oil prices plunged after two big oil exporters, Russia and Saudi Arabia, failed to agree on oil production. Then because of the rapid spread of Covid-19, stay-at-home orders and travel restrictions are issued, and global oil prices continued to fall as governments closed businesses. The international crude oil price began to recover in May 2020 thanks to the policy coordination of the Organization of the Petroleum Exporting Countries (OPEC). However, in March 2022, the oil market was stimulated by the War in Ukraine, international crude oil price have soared, requiring a full pullback in the market in the future.

There are 1906 samples in the weekly WTI price data, between January 3, 1986 and July 15, 2022, and the price trend is shown in Fig. 4 (a). Besides, from May 15, 1987 to July 15, 2022, 1836 samples of the weekly Brent benchmark are collected, and the price curve is shown in Fig. 4 (b). As for monthly data, the WTI crude oil price data includes 438 samples, which is collected from January 1986 to June 2022. Similarly, 422 price samples of monthly Brent benchmark span from May 1987 to June 2022 are gathered. The trends of these two monthly price samples can be shown in Fig. 4 (c) and Fig. 4 (d) respectively.

For the datasets used in this study, the first 80 % data is regarded as the training datasets, while the remaining 20 % is the testing datasets. The detail time span of training datasets and testing datasets and their associated statistical characteristics, including number of samples, average, median, maximum, minimum, and the standard deviation are listed in Table 3 respectively.

Table 2
Seven periods range from 1986 to 2022 [53].

Time	Representative events	Oil price fluctuations
1986–1997	Gulf War	slide
1997–2002	Asian financial crises	fluctuation
2002–2008	Strong world economic development	rise
2008–2014	War in Libya, violence escalates in Egypt	shake
2014–2022	Covid-19	plunge
2022	War in Ukraine	surge [54]

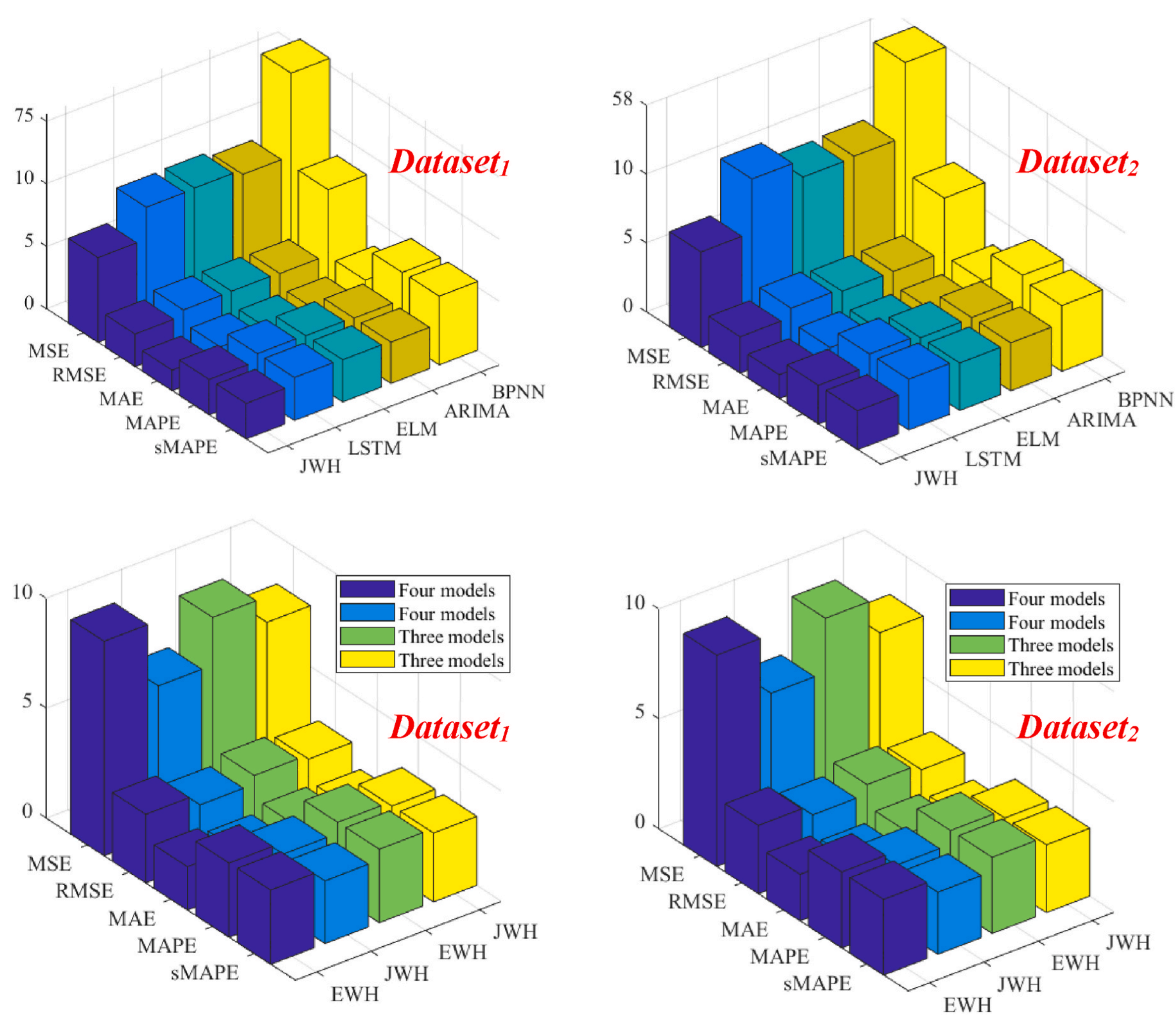

Fig. 5. Results of the different forecasting models using Dataset$_1$ and Dataset$_2$.

Table 3
Time horizon and statistical characteristics of training dataset and testing dataset.

Dataset	Division	Time horizon	Size	Mean	Median	Max	Min	Std
Dataset$_1$	**Training dataset**	1986/01/03–2015/03/27	1525	42.73	26.79	142.52	11.00	30.99
	Testing dataset	2015/04/03–2022/07/15	381	56.89	53.60	120.43	3.32	17.71
Dataset$_2$	**Training dataset**	1987/05/15–2015/07/03	1836	44.88	26.42	141.07	9.44	34.63
	Testing dataset	2015/07/10–2022/07/15	367	60.68	59.32	127.40	14.24	19.43
Dataset$_3$	**Training dataset**	1986/01–2015/03	438	42.72	26.97	133.88	11.35	30.96
	Testing dataset	2015/04–2022/07	87	56.62	54.45	114.84	15.55	17.32
Dataset$_4$	**Training dataset**	1987/05–2015/06	422	44.85	26.26	132.72	9.82	34.61
	Testing dataset	2015/07–2022/07	84	60.37	59.23	122.71	18.38	18.95

Note: Dataset$_1$: weekly WTI crude oil price data, Dataset$_2$: weekly Brent crude oil price data, Dataset$_3$: monthly WTI crude oil price data, Dataset$_4$: monthly Brent crude oil price data.

4.3. Measurement criteria

In this subsection, five metrics are chosen to compare predicting accuracy of the proposed combined and comparison models, including mean squared error (**MSE**), root mean squared error (**RMSE**), mean absolute error (**MAE**), mean absolute percentage error (**MAPE**). The error is measured in the form of a percentage by **MAPE**, with an easier presentation of the results, however, it is asymmetric, thus symmetric mean absolute percentage error (**sMAPE**) is also used in this paper.

$$MSE = \frac{1}{n}\sum_{t=1}^{n}(y_t - \hat{y}_t)^2 \tag{32}$$

$$RMSE = \left(\frac{1}{n}\sum_{t=1}^{n}(y_t - \hat{y}_t)^2\right)^{\frac{1}{2}} \tag{33}$$

$$MAE = \frac{1}{n}\sum_{t=1}^{n}|y_t - \hat{y}_t| \tag{34}$$

$$MAPE = \frac{100\%}{n}\sum_{t=1}^{n}\left|\frac{y_t - \hat{y}_t}{y_t}\right| \tag{35}$$

$$sMAPE = \frac{100\%}{n}\sum_{t=1}^{n}\left|\frac{y_t - \hat{y}_t}{(y_t + \hat{y}_t)/2}\right| \tag{36}$$

in which n is the number of testing samples, y_t represents observed value and $\hat{y}_t$ denotes predicted value.

All forecasting models are implemented on MATLAB 2019a. The related errors are measured for each experiment and their average values of ten times repeating experiments, are taken as the prediction results.

4.4. Experiment I: weekly crude oil price prediction

To verify the performance of the proposed combined model, two weekly datasets i.e., **Dataset$_1$** and **Dataset$_2$** introduced in subsection 4.2 play essential roles in this subsection. Furthermore, three comparisons are founded to highlight the superiority of the proposed combined model. First of all, the forecasting accuracy of the proposed combined model and four single models on the two datasets would be compared, reflecting the value of the combination of several models. Specifically, LSTM, ELM, ARIMA, and BPNN are selected as single models for prediction and the prediction modes of them are many-to-one, i.e. multiple historical observations are used to forecast one data. The parameter settings of the models are shown in Table A1 in **Appendix**. On the basis of these four benchmark models, the proposed time-varying weight combined model, namely **JWH** is established to obtain more stable and accurate prediction results. Besides, the proposed **JWH** would be compared with **EWH**, emphasizing the superiority of using time-varying weights for combination over using equal weights for combination. In detail, the above four benchmark models are selected to establish equal weight model and its performance on two datasets is compared with **JWH**. Furthermore, three models are selected to build **JWH$_3$** and **EWH$_3$**, whose prediction accuracy would be compared with **JWH** and **EWH** based on four models, revealing that the new combined method is less affected by poorly performing models. In this part, the prediction performance of the four benchmark models are evaluated to provide evidence for the selection of three better models, and these three models would be used to establish combined models again, then the prediction accuracy of the two strategies, i.e. combining three models and combining four models, would be compared. To quantify the prediction performance of these model, five evaluation criteria, i.e. **MSE**, **RMSE**, **MAE**, **MAPE** and **sMAPE** are calculated and the results are shown in Table 4, where the values in bold representation represent the smallest errors among the various methods.

1) **Comparison 1:** The proposed combined model is compared with four single models, which is designed to reflect the outstanding contribution of the time-varying combined model in improving the prediction accuracy. Table 4 shows the performance of different forecasting models, from which it could be observed that this novel combined model works better than benchmark models. Generally, this phenomenon can be explained that the combined model eliminates the decisive effect of the elementary model on the prediction results. Concretely, the **MSE**, **RMSE**, **MAE**, **MAPE** and **sMAPE** of **JWH** are 6.78, 2.60, 1.65, 2.83 % and 2.85 % respectively on **Dataset$_1$** while these of **JWH** are 6.93, 2.63, 1.73, 2.81 %, 2.82 % respectively on **Dataset$_2$**, which are lower than these of single models, illustrating that **JWH** outperforms all single models on the forecasting of crude oil price. From Fig. 5, we can obtain this conclusion intuitively.
2) **Comparison 2:** The prediction performance of **EWH** and **JWH** are compared based on the five evaluation metrics, asserting that assigning time-varying weights to each single models benefits to obtaining better results compared with assigning equal weights. As it shown on Table 4, we can find that the **MSE**, **RMSE**, **MAE**, **MAPE** and **sMAPE** of **JWH** are lower than these of **EWH** on both weekly datasets, which also fully illustrates the superiority of the proposed model. Results presented in Table 4 reveal the performance of the elementary methods on the entire testing datasets by enumerating the measurement metrics, which can also be shown in Fig. 5.
3) **Comparison 3: EWH** and **JWH** are designed based on three and four benchmark models respectively to show that the models presented in this paper are less influenced by the underperforming model. On the on hand, ARIMA makes the best performance on **Dataset$_1$**, while the forecast of BPNN is the worst for that it is the only one whose value of MAPE exceeds 5 % and value of MSE over 10 much more of all methods. Compared to ARIMA, the other two methods, namely LSTM and ELM, perform slightly worse on the **Dataset$_1$**, but both have much better prediction accuracy than BPNN. On the other hand, in the forecasting of weekly Brent price, the effect of ELM achieves the best, which is different from its performance on WTI. The performance of ARIMA is second only to ELM, with its MAE value only 0.03

Table 4
Forecasting performance of different models based on weekly price datasets.

Dataset	*Criteria*	Different models							
		LSTM	ELM	ARIMA	BPNN	JWH	EWH	JWH_3	EWH_3
$Dataset_1$	**MSE**	9.33	9.39	**9.00**	75.62	**6.78**	9.76	7.78	8.95
	RMSE	3.05	3.06	**3.00**	8.22	**2.60**	3.12	2.79	2.99
	MAE	2.02	1.97	**1.94**	2.92	**1.65**	1.97	1.81	1.95
	MAPE	3.41 %	3.37 %	**3.30** %	5.46 %	**2.83** %	3.37 %	3.12 %	3.33 %
	sMAPE	3.44 %	3.37 %	**3.31** %	5.49 %	**2.85** %	3.38 %	3.13 %	3.34 %
$Dataset_2$	**MSE**	10.80	**9.52**	9.61	58.57	**6.93**	9.61	7.79	9.39
	RMSE	3.29	**3.09**	3.10	7.01	**2.63**	3.10	2.79	3.06
	MAE	2.28	**2.09**	2.12	2.85	**1.73**	2.11	1.82	2.10
	MAPE	3.67 %	**3.43** %	3.47 %	5.11 %	**2.81** %	3.45 %	3.07 %	3.44 %
	sMAPE	3.69 %	**3.44** %	3.49 %	4.77 %	**2.82** %	3.45 %	3.08 %	3.46 %

Note: $Dataset_1$: weekly WTI crude oil price data, $Dataset_2$: weekly Brent crude oil price data. JWH and EWH are established by four benchmark models while JWH_3 and EWH_3 are established by three models. On each dataset, the first set of bolded data indicates the best forecasting results in four single models and the second set of bolded data suggests that the best performance can be obtained through the proposed combined method based on four models.

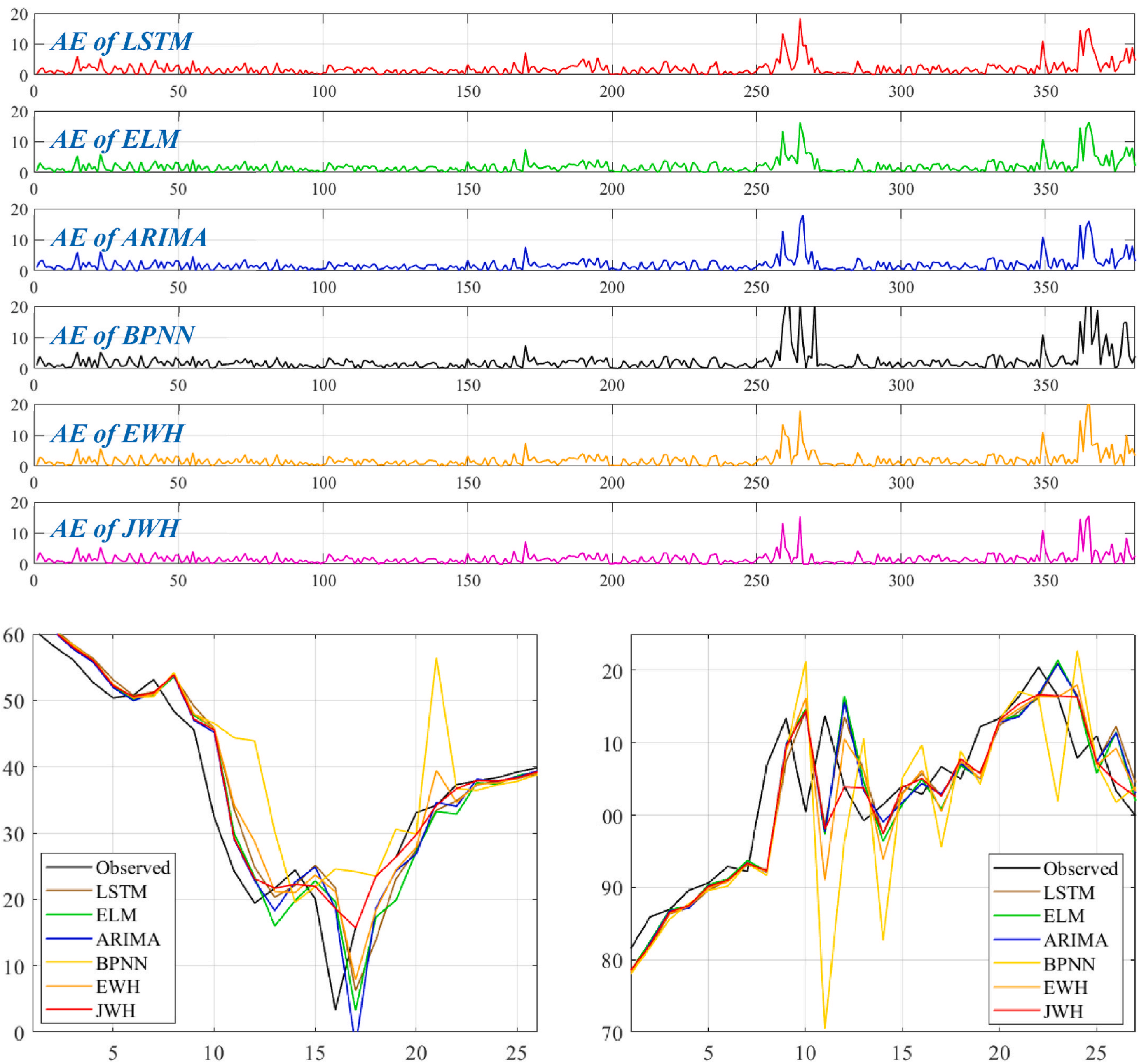

Fig. 6. The performances of different forecasting methods.

higher than ELM and its MAPE value only 0.04 % higher than ELM. LSTM performs significantly worse on **Dataset$_2$** compared to **Dataset$_1$**, which also means that its performance is more dependent on datasets. Moreover, the performance of BPNN is not as worse as its prediction of WTI crude oil price even though it is also the worst method. It is clear that for crude oil price forecasting on these two datasets, the performance and stability of BPNN is the worst, leading to its being given up. Then a new strategy with Jaynes time-varying weights would be founded by ARIMA, ELM and LSTM, whose results are shown in Table 4. Obviously, the **MSE**, **RMSE**, **MAE**, **MAPE** and **sMAPE** of **JWH$_3$** are also lower than these of **EWH$_3$**, revealing that the performance of the proposed model is always better than equal weight method in this subsection. Meanwhile, it can be found that **JWH$_3$** underperforms less than **JWH**, while **EWH$_3$** outperforms better than **EWH**, indicating the proposed method are less influenced by the underperforming model.

In general, **JWH$_3$** is the method with the highest prediction accuracy with the respect to the weekly datasets, although its effect is not as good as **JWH**. Overall, the combined prediction methods based on the variable weights of the three benchmark methods and the four benchmark methods, the performance of the latter is even better than the former. The reason for this is that even though the overall performance of BPNN on the four datasets is the worst of the four single models, it may have outperformed than other models at some sample points, making known that introducing BPNN as one of the choices of the combined model, while also incorporating the advantages of BPNN into the combined model, is beneficial to obtain a higher prediction accuracy.

The equal weight combined model **EWH$_3$** based three models performed better on **Dataset$_1$** and **Dataset$_2$** than the **EWH** based four models, this phenomenon can be explained that BPNN is the worst model of the four models and its given up would lead that the prediction accuracy becomes higher after the weight value of 1/4 belonging to the BPNN is equally assigned to the LSTM, ARIMA, and ELM.

Fig. 6 depicts the change of the absolute error (AE) in **Dataset$_1$**, which visually describes the difference between the predicted and observed values, and can be expressed mathematically as follows:

$$AE = |y_t - \widehat{y}_t| \tag{37}$$

As can be seen from the forecasts of BPNN in Fig. 6, at some of the last data points in the training set, when the raw data fluctuates greatly, the performance of BPNN is extremely bad. There are even some prominent points, that is, there are some data points with large errors in the prediction results of BPNN. Whether flat or sharply, the remaining three models perform better overall, but had slightly worse predictions at some points. As the combination of the four types of models, the prediction curves of **JWH** are relatively smooth and tend to be the observed curves.

However, it is not a coincidence that all the sub-graphs can clearly be observed that the error of several sample points is indeed very large. Due to the similarity of the results of several approaches, we only analyze the sharp points on the prediction curve of BPNN, and Table 5 presents the data points with top 5 error values and these two significantly fluctuating time periods are depicted in Fig. 6.

Table 5
Two fluctuating time periods of error curves.

Benchmark	Fluctuating time periods	Time points within top 5 AE value	Special events
Dataset$_1$	2020/3/6–2020/6/5	2020/3/27 2020/4/24 2020/5/29	**Covid-19**
	2022/3/4–2022/7/1	2022/3/25 2022/4/15	**The War in Ukraine**
Dataset$_2$	2020/2/7–2020/5/29	2020/3/13	**Covid-19**
	2022/3/4–2022/7/8	2022/3/4 2022/3/25 2022/4/15 2022/7/1	**The War in Ukraine**

Overall, for both two benchmarks, the error of BPNN is significantly increasing on the time span from February 2020 to June 2020 and March 2022 to July 2022. During these two time periods, the overall international environment is turbulent, mainly because of Covid-19 and the War in Ukraine. As one of the traditional neural networks, BPNN is obviously not suitable for the forecasting problems of time series greatly influenced by international policies and the environment. The performances of ELM, LSTM and ARIMA are significantly better than BPNN in the both two periods, but are also large volatility and have a poor fit to the observed values. This also shows that the forecasting performance is not satisfactory if one method is used to forecast crude oil price. At the same time, the performances of the combined model have a better robustness in general, which can better fit the fluctuations of crude oil price.

4.5. Experiment II: monthly crude oil price prediction

To further illustrate the validity and applicability of the developed time-varying weight combined method, the monthly price sequences for WTI crude oil and Brent crude oil are collected in this subsection. Similar to ***Experiment I***, the proposed combined method and equal weight combined model based on three single models as well as four single models are used to forecast on **Dataset$_3$** and **Dataset$_4$** respectively, and the data prediction mode is also many-to-one. The parameter settings of the models are shown in Table A1 in **Appendix.** Then all of the model evaluation criteria are shown in Table 6 and Fig. 7.

At the same time, we can also obtain three conclusions based on the three comparisons proposed in ***Experiment I***. First of all, the best method on **Dataset$_3$** and **Dataset$_4$** is the proposed model established by four benchmark models, with significantly higher prediction accuracy than the four elementary methods, showing the advantages of combining strategy. Take **MAPE** as an example, the proposed combined model achieves the best prediction results that has lowest values i.e., 5.93 % and 6.46 % respectively. Furthermore, **EWH** performs worse than **JWH** on these two datasets according to the values of **MSE**, **RMSE**, **MAE**, **MAPE** and **sMAPE**, revealing the advantages of using time-varying weights that it is more beneficial for combining the advantages of various models and so as to obtain a better prediction effect. Besides, of these four single methods, ELM performs the best with a 7.12 % **MAPE** value while LSTM performs the worst that the value of **MAPE** is 12.72 % on **Dataset$_3$** and ELM is still the model with the highest prediction accuracy while the prediction performance of BPNN is the worst on **Dataset$_4$**. Due to the lack of prediction accuracy and stability of BPNN on these four datasets, we still chose ELM, ARIMA and LSTM to reestablish combined models, and the forecasting errors are shown in the last two columns in Table 6. It can be found that among the four combined methods, **JWH** achieves the best prediction accuracy, and **JWH$_3$** is second only to **JWH**, while **EWH$_3$** who is slightly better than **EWH** ranks in the third place. The performance of these four combined methods and their performance on **Dataset$_1$** and **Dataset$_2$** are almost consistent, verifying the validity and robustness of the proposed combined strategy again.

4.6. Experiment III: partial weekly crude oil price prediction

To demonstratethe generalization ability of the established combined forecasting model deeply, two experiments would be designed based on **Dataset$_5$** and **Dataset$_6$** collected from WTI weekly crude oil price as the final illustration to verify the validity and applicability of **JWH**. As for the composition of the datasets, the first 1000 data points and the last 1000 data points of **Dataset$_1$** constitute the transport **Dataset$_5$** and **Dataset$_6$** and the first 80 % are used as training data and

Table 6
Forecasting performance of different models based on monthly price data.

Dataset	*Criteria*	Different Models							
		LSTM	ELM	ARIMA	BPNN	JWH	EWH	JWH_3	EWH_3
$Dataset_3$	**MSE**	127.03	**34.91**	36.97	106.93	**24.48**	35.63	26.86	37.01
	RMSE	11.04	**5.91**	6.08	9.97	**4.95**	5.97	5.18	6.08
	MAE	8.53	**4.33**	4.61	6.38	**3.68**	4.41	3.75	4.38
	MAPE	12.72 %	**7.12** %	7.87 %	10.69 %	**5.93** %	7.03 %	6.23	7.23 %
	sMAPE	12.98 %	**7.24** %	7.89 %	10.93 %	**6.01** %	7.17 %	6.31	7.13 %
$Dataset_4$	**MSE**	94.41	**45.13**	46.05	122.72	**31.72**	42.57	37.25	45.98
	RMSE	9.64	**6.71**	6.79	10.67	**5.63**	6.52	6.10	6.78
	MAE	7.12	**4.87**	4.96	6.83	**4.10**	4.68	4.38	4.81
	MAPE	10.47 %	**7.68** %	8.05 %	10.84 %	**6.46** %	7.37 %	7.07 %	7.60 %
	sMAPE	10.79 %	**7.86** %	8.15 %	10.86 %	**6.58**	7.54 %	7.21 %	7.80 %

Note: $Dataset_3$: monthly WTI crude oil price data, $Dataset_4$: monthly Brent crude oil price data. JWH and EWH are established by four benchmark models while JWH_3 and EWH_3 are established by three models. On each dataset, the first set of bolded data indicates the best forecasting results in four single models and the second set of bolded data suggests that the best performance can be obtained through the proposed combined model based on four models.

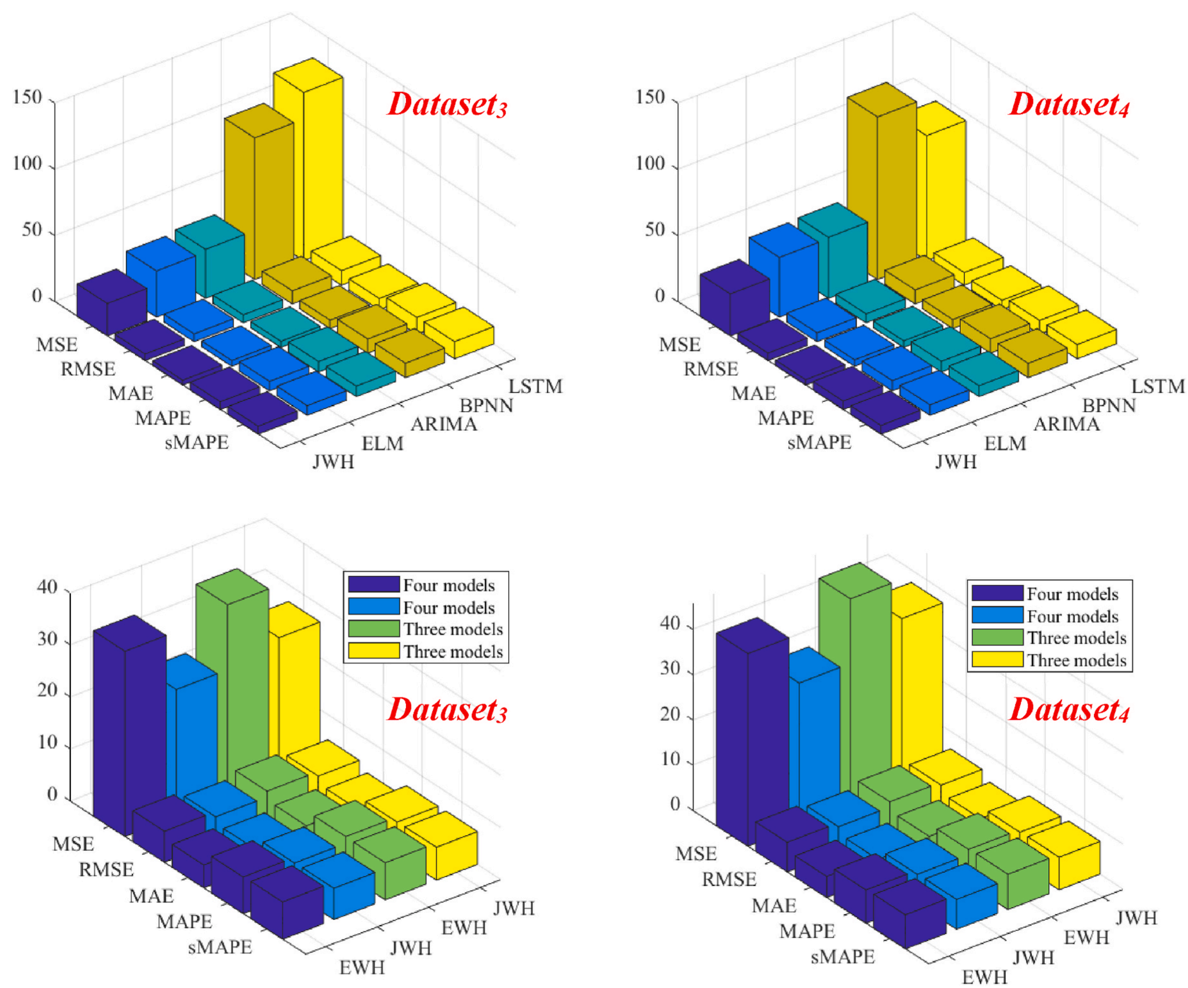

Fig. 7. Results of the different forecasting models using $Dataset_3$ and $Dataset_4$.

the remaining 20 % are used as test data respectively. The forecasting errors of **JWH** and all compared models are shown in Table 7. The same conclusion can be drawn as ***Experiment I*** and ***Experiment II*** by comparing the size of the relevant evaluation indexes in Table 7, namely, **JWH** performs the best in all models, indicating the superiority and applicability of the proposed model again.

5. Discussion

From the performance on the datasets, five measurement criteria have validated that the forecasting accuracy of this combined model is always higher than that of four benchmark methods and equal weight hybrid model. To test the advanced predictive power of this combined method, DM test introduced by Diebold and Mariano [55], one of hypothesis testing methods, can be used to judge whether the two prediction models are significantly different at a given confidence degree.

To comprehensively compare the performance of two methods, DM test is carried out based on two loss functions, i.e., L1 (mean absolute deviation, **MAD**) and L2 (mean square error, **MSE**). Since ***Experiment III*** is designed based on part of the weekly data of the WTI crude oil price,

Table 7
Forecasting performance based on partial weekly WTI crude oil price.

Dataset	*Criteria*	Different Models							
		LSTM	ELM	ARIMA	BPNN	JWH	EWH	JWH_3	EWH_3
$Dataset_5$	**MSE**	78.04	**18.78**	19.55	100.48	**4.84**	26.71	5.58	8.82
	RMSE	8.22	**4.07**	4.10	9.66	**2.20**	5.17	2.36	2.97
	MAE	4.47	**2.37**	2.38	5.14	**1.54**	3.11	1.61	1.94
	MAPE	16.86	**8.92**	8.92	19.50	**5.33**	11.65	5.66	6.97
	sMAPE	15.64	**8.79**	8.75	17.94	**5.39**	11.58	5.68	6.98
$Dataset_6$	**MSE**	65.44	15.66	**15.07**	83.48	**13.76**	17.59	13.98	16.59
	RMSE	7.78	3.95	**3.88**	8.77	**3.71**	4.19	3.74	4.07
	MAE	3.47	2.47	**2.44**	3.78	**2.37**	2.51	2.37	2.43
	MAPE	5.66	3.79	**3.76**	6.54	**3.67**	4.01	3.67	3.89
	sMAPE	5.77	3.80	**3.78**	6.27	**3.70**	4.03	3.71	3.91

Note: $Dataset_5$: the first 1000 data points of weekly WTI crude oil price, $Dataset_6$: the last 1000 data points of weekly WTI crude oil price. JWH and EWH are established by four benchmark models while JWH_3 and EWH_3 are established by three models. On each dataset, the first set of bolded data indicates the best forecasting results in four single models and the second set of bolded data suggests that the best performance can be obtained through the proposed combined model based on four models.

we performed DM test on four complete datasets and the results of DM test by comparing the combined method and other methods are listed in Table 8 respectively.

The forecasting sequences of the novel combined method are compared with these of single models and equal weights hybrid model by DM test. Assume that the initial assumption H_0 is: there is no significant local difference between the prediction ability of the two methods.

Table 8 reveals that the values are always greater than 0 and the significance levels are always below 5 % regardless of whatever the compared model is and whatever the loss function is selected. Then the initial assumption H_0 is rejected, leading a conclusion that there are significant local differences between the performances of the two forecasting models. The values which are always bigger than 0, demonstrating that the novel combined method is outperformed the benchmark methods and the equal weight combined model.

To compare the proposed method with other methods in previous studies, we have collected and summarized some methods in other literature, and compared with our prediction results roughly and the **MAPE** values are listed as follows (see Table 9).

In this paper, the MAPE values of the proposed time-varying model is 2.83 %, 2.81 % on **$Dataset_1$** and **$Dataset_2$,** respectively. It is obviously that on other weekly datasets, the prediction accuracy of the methods in

Table 8
The results of DM test with different loss functions.

Dataset	Compared model	MAD		MSE	
		value	*p*-value	value	*p*-value
$Dataset_1$	LSTM	6.27	1.01e-9	4.22	3.12e-5
	ELM	4.95	1.14e-6	3.58	3.85e-4
	ARIMA	4.36	1.65e-5	2.35	1.95e-2
	BPNN	4.80	2.31e-6	2.65	8.41e-3
	EWH	6.63	1.13e-10	3.49	5.49e-4
$Dataset_2$	LSTM	7.53	4.08e-13	5.22	3.06e-7
	ELM	6.10	2.66e-9	3.26	1.23e-3
	ARIMA	6.10	2.70e-9	3.56	4.26e-4
	BPNN	5.60	4.15e-8	2.93	3.64e-3
	EWH	9.02	1.11e-17	4.91	1.35e-6
$Dataset_3$	LSTM	7.36	1.00e-10	4.71	9.53e-6
	ELM	2.32	2.25e-2	2.80	6.25e-3
	ARIMA	2.39	1.88e-2	2.14	3.55e-2
	BPNN	3.64	4.62e-4	3.05	3.04e-3
	EWH	4.33	4.06e-5	3.33	1.28e-3
$Dataset_4$	LSTM	5.29	9.66e-7	3.31	1.38e-3
	ELM	2.46	1.61e-2	2.25	2.72e-2
	ARIMA	2.31	2.34e-2	2.06	4.26e-2
	BPNN	2.67	9.25e-3	2.56	1.22e-2
	EWH	3.11	2.59e-3	2.77	6.86e-3

Note: $Dataset_1$: weekly WTI crude oil price data, $Dataset_2$: weekly Brent crude oil price data, $Dataset_3$: monthly WTI crude oil price data, $Dataset_4$: monthly Brent crude oil price data.

Table 9
MAPE values on weekly time series forecasting.

Authors	Dataset	MAPE
[56].	Weekly futures price of the carbon in the EU emission trading system from 2008 to 2019	4.21 %
[57]	Weekly data of four highly rich solar radiation regions in Queensland, Australia from 1905 to 2018	5.55 %
[58]	Weekly World health organization covid cumulative dataset form January 20, 2020 to September 18, 2020	12.47 %
[59].	Weekly Newcastle steam coal spot price (NPCSP) from 2014 to 2022	6.03 %
[60]	Weekly reported dengue cases obtained in the Singapore Ministry from 2012 to 2022	18.5 %

these literature is lower than that of the time-varying weight combined model, which verifies the predictive ability and robust performance of the proposed combined model again.

6. Conclusion

The study on crude oil price prediction is of great significance. For example, the cost of energy can be directly affected by crude oil price, then investment decisions are influenced in turn, so precise forecasts of crude oil price is beneficial to make investment decisions [3]. Besides, the supply of crude oil is affected by the international situation, geographical environment, transportation conditions, etc. The study of crude oil price forecasting model is helpful for the government to make decisions, so as to alleviate the national energy security problems [4].

For the reason that traditional statistical models have a poor ability to fit nonlinear trends and classical neural networks and deep learning models have a strong dependence on the selection of hyper parameters, the crude oil price cannot always be accurately predicted using single models, especially when the price fluctuates strongly. In contrast, establishing combined models based on several single models can greatly reduce the risk of unreasonable selection of models, and at the same time, each model is given a certain weight, which is more helpful to obtain higher prediction accuracy. Then how to determine the weights of different single models becomes particularly important.

In this work, to predict crude oil price with characteristics of historical information, LSTM, ELM, ARIMA and BPNN are introduced to establish a novel combined model whose weights are calculated by Shannon information entropy and Jaynes maximum entropy.

The forecasting performance of the novel combined method is tested by two international benchmarks, namely, WTI price sequences and Brent price sequences. The conclusions can be given as follows.

1) The combined method proposed in this research achieves the highest prediction accuracy compared to four benchmark models. The MAPE

of **JWH** on four datasets are 2.83 %, 2.81 %, 5.93 %, 6.46 %, respectively while the MAPE of the single model with best performance are 3.30 %, 3.43 %, 7.12 %, 7.68 %, respectively, showing that the performance of this method achieves the best performance, whether using the weekly data or the monthly data.
2) Compared with the commonly used equal weight combined models, the prediction performance is slightly better. At the same time, two strategies with three model combinations and four models combinations are designed and the **JWH** based on four benchmark models achieves the best, indicating that the new combined method is less affected by poorly performing models.
3) By analyzing the prediction performance of several models when an emergency such as Covid-19 and the War in Ukraine occurs (**Experiment *I***), it can be found that the novel combined method has an advantage in fitting the data with strong volatility, that is, the model is relatively robust.
4) ***Experiments III*** on two additional datasets obtained by the decomposition of WTI weekly crude oil price sequence shows the highest prediction accuracy of **JWH,** which further reveals the validity and applicability of the proposed crude oil price forecasting model.

In general, the work can be applied to crude oil price forecasting with better forecasts to assist the government and economic workers to avoid risks, however, the model still has some limitations. First, only the historical price is used to forecast in this study, but in fact, the influential external factors such as exchange rate and political change can be taken into account. To further enhance prediction performance, entering the crude oil price forecasting models with multiple variables is the future work direction, which requires us to collect more external data and use it as the inputs to the forecasting model. Second, the single models adopted in this study are representative methods in their respective fields where many similar models within. Perhaps there are several alternative models, whose forecasting accuracy are higher than the models adopted in this work, thus one aim of our future research work is to explore novel useful deep learning methods and establish multiple different forms of the combined models. In summary, the massive data characteristics and complex variable factors brought by the big data era as well as the continuous development of new deep learning models deserve to be explored in future research to enhance the interpretability and improve the prediction performance.

CRediT authorship contribution statement

Longlong Liu: Writing – review & editing, Supervision. **Suyu Zhou:** Writing – original draft, Software, Methodology, Investigation. **Qian Jie:** Writing – review & editing. **Pei Du:** Writing – review & editing, Supervision, Resources, Conceptualization. **Yan Xu:** Writing – review & editing, Supervision. **Jianzhou Wang:** Supervision.

Declaration of competing interest

We declare that we have no financial and personal relationships with other people or organizations that can inappropriately influence our work. This manuscript is our own work and the content of this paper has not been copied from elsewhere. This manuscript has not been published before nor submitted to another journal for the consideration of publication and all data measurements are genuine results and have not been manipulated. In addition, none of the authors have any financial or scientific conflicts of interest with regard to the research described in this manuscript.

Data availability

Data will be made available on request.

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Appendix

Table A1
Setting of the experimental parameters.

Model	Experimental parameter	Default value	
		Weekly experiments	Monthly experiments
BPNN	Training function	Trainlm	Trainlm
	Transfer function	Sigmoid	Sigmoid
	Neuron number in the input layer	6	4
	Neuron number in the hidden layer	12	12
	Neuron number in the output layer	1	1
	Learning rate	0.01	0.01
	Maximum number of iteration	2000	2000
ELM	Transfer function	Sigmoid	Sigmoid
	Neuron number in the input layer	6	4
	Neuron number in the hidden layer	20	20
	Neuron number in the output layer	1	1
LSTM	Training algorithm	Adam	Adam
	Maximum number of iteration	2000	2000
	Initial learning rate	0.003	0.003
	Learn rate schedule	Piecewise	Piecewise
	Gradient threshold	0.9	0.9
	Learn rate drop period	150	150
	Learn rate drop factor	0.15	0.15
	Neuron number in the input layer	1	1
	Neuron number in the hidden layer	200	200
	Neuron number in the output layer	1	1

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